

Applied Statistical Analysis I

Multiple linear regression

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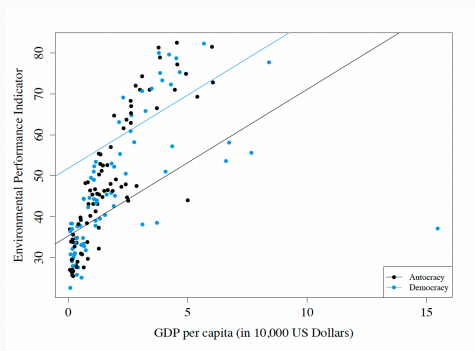
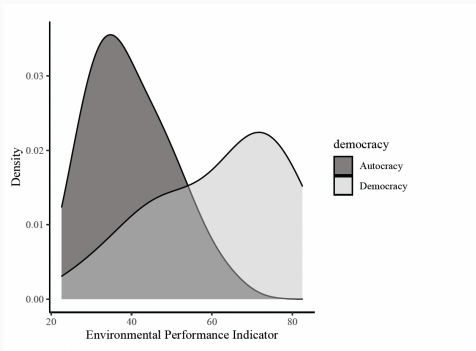
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Categorical Independent Variables

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Categorical IVs: Multiple Categories

$$\text{Environmental Performance}_i = \alpha + \beta_1 \times \text{Income} + \beta_2 \times \text{Region}_i + \epsilon_i$$

```
## table(qog_data$ht_region)
##
##           Eastern Europe (1)           Latin America(2)
##                28                20
## North Africa & the Middle East (3) Sub-Saharan Africa (4)
##                20                49
## Western Europe and North America (5)           East Asia (6)
##                27                6
##           South-East Asia (7)           South Asia (8)
##                11                8
##           The Pacific (9)           The Caribbean (10)
##                12                13
```

Categorical IVs: in R

```
1 # Load package
2 library(fastDummies)
3
4 # Create dummy variables for categorical variable
5 qog_data <- dummy_cols(qog_data,
6                         select_columns = c("ht_region"))
7
8 # Print first 5 rows in dataset
9 head(qog_data[c("ht_region_1",
10                "ht_region_2",
11                "ht_region_3",
12                "ht_region_4",
13                "ht_region_5",
14                "ht_region_6",
15                "ht_region_7",
16                "ht_region_8",
17                "ht_region_9",
18                "ht_region_10")], 5)
```

```
## ht_region_1 ht_region_2 ht_region_3 ht_region_4 ht_region_5
## 1          0          0          0          0          0
## 2          1          0          0          0          0
## 3          0          0          1          0          0
## 4          0          0          0          0          1
## 5          0          0          0          1          0
## ht_region_6 ht_region_7 ht_region_8 ht_region_9 ht_region_10
## 1          0          0          1          0          0
## 2          0          0          0          0          0
## 3          0          0          0          0          0
## 4          0          0          0          0          0
## 5          0          0          0          0          0
```

Categorical IVs: in R

```
1 # Run regression model
2 m2 <- lm(epi_epi ~ income +
3         ht_region_1 + ht_region_2 + ht_region_3 +
4         # no region 4 (Sub-Saharan Africa) = reference category.
5         ht_region_5 + ht_region_6 + ht_region_7 + ht_region_8 + ht_region_9 +
6         ht_region_10, data = qog_data)
7
8 # Print results
9 summary(m2)
```

Coefficients: (1 not defined because of singularities)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	32.3992	1.1296	28.683	< 2e-16 ***
income	1.7410	0.4061	4.287	3.23e-05 ***
ht_region_1	18.4245	1.8769	9.817	< 2e-16 ***
ht_region_2	11.6208	2.0362	5.707	6.01e-08 ***
ht_region_3	9.4434	2.4665	3.829	0.000189 ***
ht_region_5	35.2532	2.4854	14.184	< 2e-16 ***
ht_region_6	16.2287	3.6737	4.418	1.91e-05 ***
ht_region_7	4.1247	2.7820	1.483	0.140281
ht_region_8	-2.1694	3.2676	-0.664	0.507774
ht_region_9	NA	NA	NA	NA
ht_region_10	11.0665	3.5607	3.108	0.002257 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.528 on 149 degrees of freedom
(35 observations deleted due to missingness)

Multiple R-squared: 0.7897, Adjusted R-squared: 0.777
F-statistic: 62.16 on 9 and 149 DF, p-value: < 2.2e-16

Categorical IVs: in R

```
1 # Use relevel to code dummy variables on the fly
2 # specify region 4 (Sub-Saharan Africa) = reference category
3 m3 <- lm(eps_1 ~ income + relevel(as.factor(ht_region), ref = "4"),
4         data = qog_data)
5
6 # Print results
7 summary(m3)
```

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relevel(as.factor(ht_region), ref = "4")3	9.4434	2.4665	3.829	0.000189 ***
relevel(as.factor(ht_region), ref = "4")5	35.2532	2.4854	14.184	< 2e-16 ***
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relevel(as.factor(ht_region), ref = "4")8	-2.1694	3.2676	-0.664	0.507774
relevel(as.factor(ht_region), ref = "4")10	11.0665	3.5607	3.108	0.002257 **

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Under control of income, Eastern Europe has an Environmental Performance Index score of 18.4245 scale points higher than Sub-Saharan Africa.

Using Dummy Variables for Multiple Categories

- Dummy variable trap:
 - Base group is represented by the intercept.
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 - Base group is represented by the intercept.
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- **Solution:** Use $k - 1$ dummies
- Interpretation always **relative to baseline**.

Social Class	Dummy Variables		
	D_1	D_2	
lower	0	0	$\hat{Y} = \hat{\beta}_0$
middle	1	0	$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1$
upper	0	1	$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_2$

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- When estimating $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1\tilde{D}_1 + \hat{\beta}_2\tilde{D}_2$ then the estimated coefficient of the second dummy, $\hat{\beta}_2$, represents (by design!) the difference between middle and upper class.

Interactions

Modeling Interactions

- So far, we have only been adding variables in an additive manner:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \epsilon$$

- Suppose we want to test a hypothesis that the relation between an independent variable X_i and a dependent variable Y depends on the value of another dummy variable D .
- Think of: $\text{Income} = \beta_0 + \beta_1 \text{Education} + \beta_2 \text{Female} + \beta_3 \text{Education} \times \text{Female} + \epsilon$
- The effect of X_1 on Y is also called **conditional** because the hypothesized effect is conditional on D .
- In other words, if D is 1, the relation between X_1 and Y is different than when D is zero.
- This is what we also call an **interaction effect**.

Modeling Interactions: Interpretation

- An interaction effect **conditions** the effect of an independent variable (e.g., Education) on the dependent variable (e.g., Income).
- Interaction model if $D = 0$ (condition is absent):

$$Y = \beta_0 + \beta_1 \cdot X_1 + \beta_2 \cdot 0 + \beta_3 \cdot X_1 \cdot 0 + \epsilon = \beta_0 + \beta_1 X_1 + \epsilon$$

- Interaction model if $D = 1$ (condition is present):

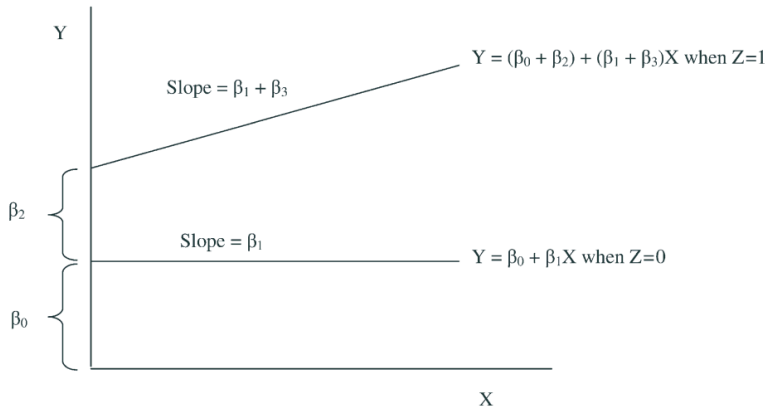
$$Y = \beta_0 + \beta_1 \cdot X_1 + \beta_2 \cdot 1 + \beta_3 \cdot X_1 \cdot 1 + \epsilon = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)X_1 + \epsilon$$

- In other words, we get an **intercept shift** and a **change in slopes**.
- Do not interpret constitutive terms (i.e., $\hat{\beta}_1$ and $\hat{\beta}_2$ as if they are unconditional effects!)

Modeling Interactions: Interpretation

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ + \varepsilon$$

Hypothesis H_1 : An increase in X is associated with an increase in Y when condition Z is met, but not when condition Z is absent.



Modeling Interactions: Interpretation

$$\hat{Y}_i = \beta_0 + \beta_1 X_i + \beta_2 M_i + \beta_3 (X_i M_i) + \epsilon_i$$

The interpretation of the regression coefficients changes:

- β_0 is the expected value of Y when $X = 0$ and $M = 0$
- β_1 is the average change in Y when X increases by one unit, when $M = 0$.
- β_2 is the average change in Y when M increases by one unit, when $X = 0$.
- β_3 is the *interaction term* of X and M .

How much X impacts the
change in Y based on M

If we rearrange the terms, we get:

$$\hat{Y}_i = \beta_0 + \beta_2 M_i + (\beta_1 + \beta_3 M_i) X_i + \epsilon_i$$

β_3 is the added increase in β_1 , if M increases by one unit.

Modeling Interactions with Continuous Variables

- So far we studied interactions with dummy variables.
- But, we can interact **continuous variables** as well.
- Assume instead of a dummy D , X_2 to be continuous.
- Example: *Close (in time) presidential elections will reduce the effective number of electoral parties in subsequent parliamentary elections if and only if the number of presidential candidates is sufficiently low.*
- Thus,

$$\begin{aligned}\text{Electoral Parties} = & \beta_0 + \beta_1 \text{Proximity} + \beta_2 \text{Presidential Candidates} \\ & + \beta_3 \text{Proximity} \times \text{Presidential Candidates} + \epsilon\end{aligned}$$

Modeling Interactions with Continuous Variables

- In this case, the effect of the explanatory variable on the outcome variable **gradually changes** as another variable changes.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \dots + \epsilon$$

$$Y = \beta_0 + \beta_2 X_2 + (\beta_1 + \beta_3 X_2) \cdot X_1 + \epsilon$$

- **Marginal effect** of X_1 on Y represents the effect of change in X_1 on the expected change in Y , especially when the change in the independent variable (X_1) is infinitely small (marginal).

Modeling Interactions with Continuous Variables

Table 1 The impact of presidential elections on the effective number of electoral parties. Dependent variable: Effective number of electoral parties

<i>Regressor</i>	<i>Model</i>
Proximity	-3.44** (0.49)
PresidentialCandidates	0.29* (0.07)
Proximity*PresidentialCandidates	0.82** (0.22)
Controls	—
Constant	3.01** (0.33)
R^2	0.34
N	522

* $p < 0.05$; ** $p < 0.01$ (two-tailed). Control variables not shown here.
Robust standard errors clustered by country in parentheses.

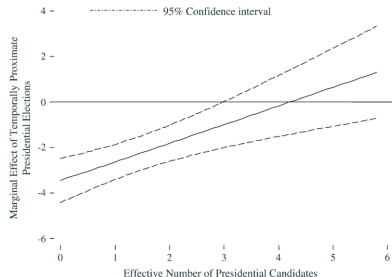


Fig. 3 The marginal effect of temporally proximate presidential elections on the effective number of electoral parties.

The marginal effect of proximity on the effective number of electoral parties is -2.62 point for every extra presidential candidate.

In-class example:

$$\text{Environmental Performance}_i = \alpha + \beta_1 \text{Income}_i + \beta_2 \text{Regime Type}_i + \beta_3 \text{Income}_i * \text{Regime Type}_i + \epsilon_i$$

```
1 # Run regression model with interaction term
2 int_m2 <- lm(emi_epi ~ income + democracy + income*democracy, data = qog_data)
3
4 # Print results
5 summary(int_m2)
```

```
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    37.1474     1.0684  34.768 < 2e-16 ***
## income         2.1902     0.4532   4.833 3.24e-06 ***
## democracyDemocracy 3.4490     2.7819   1.240  0.217
## income:democracyDemocracy 5.1029     0.8686   5.875 2.55e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 9.046 on 153 degrees of freedom
## (37 observations deleted due to missingness)
## Multiple R-squared:  0.6879, Adjusted R-squared:  0.6818
## F-statistic: 112.4 on 3 and 153 DF, p-value: < 2.2e-16
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```

- The average Environmental Protection Index (EPI) for poor (Income = 0) autocracies is 37.1474 scale points (α).
- For autocracies, with every additional 10,000 USD of income, the EPI increases by 2.1902 scale points, on average (β_1) → Income effect for autocracies.
- For poor democracies, the EPI is 3.4490 scale points higher on average, in comparison to poor autocracies (β_2).
- For democracies, with every additional 10,000 USD of income, the EPI increases by 7.2931 scale points ($\beta_1 + \beta_3 = 2.1902 + 5.1029 = 7.2931$) → Income effect for democracies.

The systematic components:

Model for Autocracies (*democracy* = 0):

$$\hat{Y}_i = 37.1474 + (2.1902 \cdot \text{Income}_i) + (3.4490 \cdot \text{Regime Type}_i) \\ + (5.1029 \cdot \text{Income}_i \cdot \text{Regime Type}_i)$$

$$\hat{Y}_i = 37.1474 + (2.1902 \cdot \text{Income}_i) + (3.4490 \cdot 0) + (5.1029 \cdot \text{Income}_i \cdot 0)$$

$$\hat{Y}_i = 37.1474 + 2.1902 \text{Income}_i$$

Model for Democracies (*democracy* = 1)

$$\hat{Y}_i = 37.1474 + (2.1902 \cdot \text{Income}_i) + (3.4490 \cdot \text{Regime Type}_i) \\ + (5.1029 \cdot \text{Income}_i \cdot \text{Regime Type}_i)$$

$$\hat{Y}_i = 37.1474 + (2.1902 \cdot \text{Income}_i) + (3.4490 \cdot 1) + (5.1029 \cdot \text{Income}_i \cdot 1)$$

$$\hat{Y}_i = 40.5964 + (7.2931 \cdot \text{Income}_i)$$

Visualization of example:

